



Topic wise preparation plan +cS +DSA+6 Months

Here's a **topic-wise, 6-month integrated plan** for mastering **CS core subjects + DSA + projects**, optimized for placement preparation. [\[1\]](#) [\[2\]](#) [\[3\]](#) [\[4\]](#) [\[5\]](#) [\[6\]](#)

Month 1: Programming & Core CS Foundations

Goals: Build strong base in coding language, start DSA, begin key CS subjects.

Topics:

- **Programming Language:** Syntax, loops, conditionals, recursion, OOP concepts (C++/Java/Python)
- **DSA:** Arrays, Strings, Time & Space Complexity, Recursion basics
- **CS Subjects:**
 - **Operating Systems:** Process management, scheduling, threads
 - **DBMS:** SQL basics, normalization, ACID properties
- **Practice Platforms:** HackerRank, GfG basic problems

Project: Simple CLI app (e.g., student record system using file handling)

Month 2: Fundamental DSA & Core Integration

Goals: Develop logical problem-solving skills and dive deeper into CS concepts.

Topics:

- **DSA:** Linked Lists, Stacks, Queues, Searching & Sorting algorithms
- **CS Subjects:**
 - **Computer Networks:** OSI layers, IP addressing, TCP vs UDP
 - **OS:** Deadlocks, synchronization, paging vs segmentation
- **SQL Practice:** Joins, subqueries, constraints
- **Practice:** 2–3 problems daily (LeetCode Easy/Medium)

Project: CRUD web app or basic portfolio site

Month 3: Advanced DSA & DBMS Deep Dive

Goals: Tackle moderate DSA problems and link theory with practice.

Topics:

- **DSA:** Trees, BSTs, Hashing, Sliding Window, Two Pointers
- **CS Subjects:**
 - **DBMS:** Transactions, indexing, query optimization
 - **OOPs:** Abstraction, inheritance, polymorphism with examples
- **Mock Tests:** Basic coding + aptitude practice
- **Practice:** 3–4 problems daily + weekly mock test

Project: Database-based app (library management system using SQL backend)

Month 4: Complex DSA & Project Development

Goals: Master core algorithms and start major development work.

Topics:

- **DSA:** Graphs (BFS, DFS, shortest paths, MST), Recursion backtracking
- **Core CS (Revision):** OS & CN recaps with MCQs
- **Development:** Work on a full-stack or mobile project (React, Node.js, Django, or Flutter)
- **Problem Solving:** 150–200 questions cumulative target [\[4\]](#) [\[7\]](#) [\[1\]](#)

Project: E-commerce or task-tracking app with authentication

Month 5: Dynamic Programming, Optimization & Aptitude

Goals: Master advanced problem-solving and begin interview preparation.

Topics:

- **DSA:** Dynamic Programming patterns (knapsack, LIS, partitions)
- **Aptitude:** Quantitative (Time, Work, Probability)
- **CS Subjects:** Quick revision of OS, DBMS, CN
- **Mock Interviews:** Peer/practice sessions twice per week

Project: Refine project — add deployment (GitHub Pages, Render, or AWS)

Month 6: Revision, System Design & Mock Interviews

Goals: Final polish for placements and simulate real tests.

Topics:

- **DSA:** Mixed topic practice (random sets from Arrays, Graphs, DP)
- **System Design (Basics):** Scalability, caching, load balancing
- **Mock Rounds:** Conduct 3–5 mock interviews + complete topic-wise revisions
- **Resume & Portfolio:** Highlight projects and GitHub contributions^[5] ^[6]

Project: Final documentation, demo video, and deployment readiness

Weekly Schedule (Recommended)

Day	Focus
Mon–Wed	DSA (new topics + problem-solving)
Thu	CS Subject (OS/DBMS/CN)
Fri	Project updates/features
Sat	Revision or practice test
Sun	Resume review + rest/light coding

Key Outcomes After 6 Months

- **DSA Mastery:** 250–300 problems solved across LeetCode, GfG
- **CS Expert:** Strong fundamentals in OS, DBMS, CN, OOPs
- **Project Portfolio:** 2–3 deployable projects
- **Interview-Ready:** Clear on concepts + strong resume narrative

This topic-wise plan ensures balanced skill development—**theory, coding, and practical exposure**—matching the expectations of both **product-based** and **service-based** recruiters.^[2]
^[6] ^[1] ^[4] ^[5]



1. https://www.reddit.com/r/leetcode/comments/1hmw5lq/my_6month_dsa_plan_deep_dive_conquer/
2. <https://www.lets-code.co.in/articles/placement/>
3. <https://www.geeksforgeeks.org/blogs/prepare-cs-core-subjects-for-placements/>
4. <https://dev.to/kartikdevsharma/how-to-prepare-for-dsa-i8f>
5. <https://www.scribd.com/document/839743911/6-Months-Placement-Roadmap>
6. <https://www.scribd.com/document/885720656/Complete-Computer-Science-6-Month-Plan>
7. https://www.reddit.com/r/Btechtards/comments/1bniob6/complete_competitive_programming_dsa_guide_that_i/

8. https://www.reddit.com/r/redtaganna/comments/1fkdneyi/concise_dsa_guide_your_path_to_coding_interviews/
9. https://www.reddit.com/r/Btechtards/comments/1hwloqe/placement_coming_up_in_6_months_and_want_to_crack/
10. <https://www.geeksforgeeks.org/complete-roadmap-to-learn-dsa-from-scratch/>